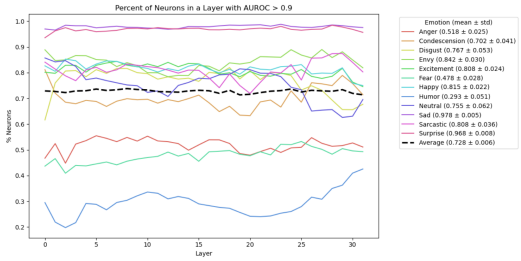

Figure 9: Results of ML-AURA by layer and emotion for LLama3.1-8B-Base (also in main paper, but reproduced here for ease of comparison with Llama3.1-8B-Instruct). Results are in terms of percent of neurons with an AUROC score above 0.9.

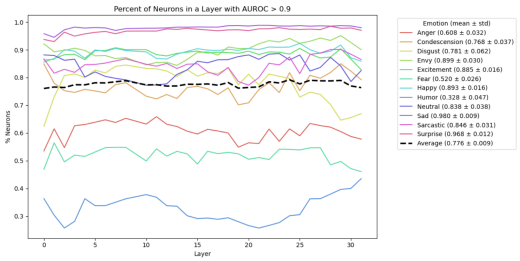

Figure 10: Results of ML-AURA by layer and emotion for LLama3.1-8B-Instruct. Results are in terms of percent of neurons with an AUROC score above 0.9.

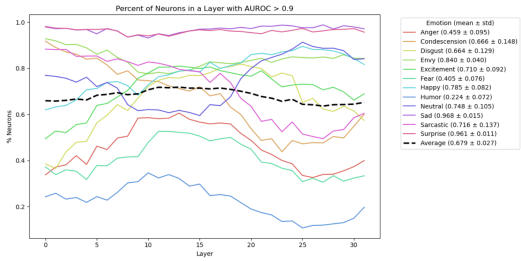

Figure 11: Results of ML-AURA by layer and emotion for Olmov2-Base. Results are in terms of percent of neurons with an AUROC score above 0.9.

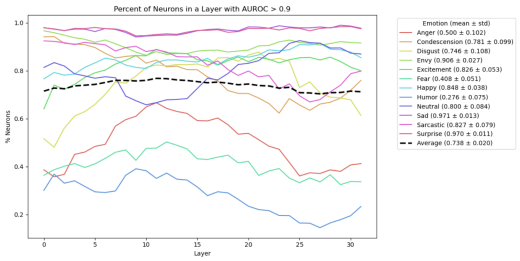

Figure 12: Results of ML-AURA by layer and emotion for Olmov2-Instruct. Results are in terms of percent of neurons with an AUROC score above 0.9.

F ML-AURA Across Models

The ML-AURA analysis in Section 4.3 was reproduced with Llama3.1-8B-Instruct, Olmov2-Base OLMo et al. (2024) (11), Olmov2-Instruct (12), and Ministral-8B (AI, 2024) (13) showing results consist with what was found with Llama-3.1. Olmov2-Base performed worse than Olmov2-Instruct. Both of those models performed worse than Ministral-8B and the Llama at the same stage of training. The neurons in the Ministral-8B model were only slightly able than the neurons in the Llama model to separate between emotions. However, in all models more than the majority of neurons at all layers are able to do 1vAll classification of the specified emotion of interest at an AUROC > 0.9 showing great separability in how the different emotions are represented with a low amount of confusion.

Instruction-tuning was found to improve neuron's performance on separating emotion. For Llama-3.1-8B, instruction-tuning gave an average 4.8% boost in the number of neurons per layer that were able to separate emotions.

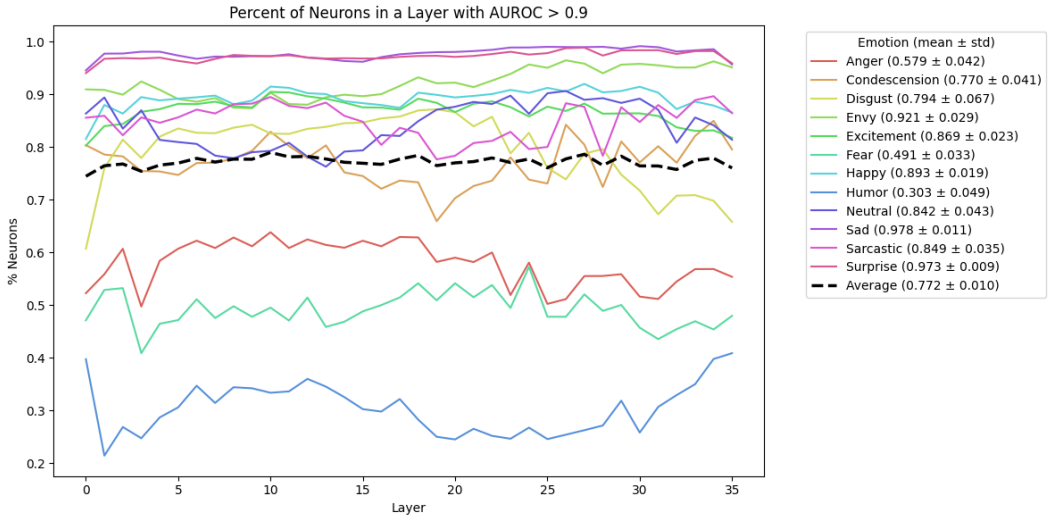

Figure 13: Results of ML-AURA by layer and emotion for Ministral. Results are in terms of percent of neurons with an AUROC score above 0.9.

G Emotional Consistency Across Principal Axes

Model	PC	Avg. Spearman	Avg. Kendall
Llama3.1-8B-Base	PC1	0.87	0.82
	PC2	0.83	0.77
	PC3	0.80	0.74
Llama3.1-8B-Instruct	PC1	0.87	0.81
	PC2	0.92	0.85
	PC3	0.84	0.78
Olmov2-Base	PC1	0.76	0.71
	PC2	0.70	0.65
	PC3	0.69	0.64
Olmov2-Instruct	PC1	0.84	0.77
	PC2	0.74	0.69
	PC3	0.70	0.65
Ministral	PC1	0.94	0.91
	PC2	0.83	0.83
	PC3	0.79	0.79

(a) Synthetic emotion dataset.

Model	PC	Avg. Spearman	Avg. Kendall
Llama3.1-8B-Base	PC1	0.92	0.86
	PC2	0.74	0.68
	PC3	0.73	0.68
Llama3.1-8B-Instruct	PC1	0.98	0.98
	PC2	0.85	0.78
	PC3	0.70	0.65
Olmov2-Base	PC1	0.76	0.70
	PC2	0.69	0.65
	PC3	0.68	0.64
Olmov2-Instruct	PC1	0.83	0.77
	PC2	0.74	0.69
	PC3	0.70	0.65
Ministral	PC1	0.93	0.90
	PC2	0.74	0.68
	PC3	0.71	0.66

(b) Go-emotions dataset.

Table 6: Correlation between emotion rankings in latent space across model layers.

The emotional space is strikingly stable across models, especially along the first three principal components. Table 6 reports average Spearman and Kendall correlations of emotion orderings across layers for each model, showing consistently high values for both the synthetic dataset and the more fine-grained Go-Emotions labels. This suggests that the models' emotional manifolds possess stable, interpretable axes.

This structure is not only stable within models, but also consistent across them in terms of relative emotion positioning as shown in Table 7. The high rank-order correlations suggest that the emotional geometry described later in this section reflects a shared conceptual structure across models. However, results from the inter-model alignment analysis indicate that these shared structures are embedded in distinct internal coordinate systems, requiring high-complexity transformations to align. Thus, while the emotional manifolds are topologically consistent, their parameterizations remain model-specific—likely shaped by architectural and pretraining differences.

Model	PC	Avg. Spearman	Avg. Kendall
Llama3.1-8B-Instruct	PC1	0.73	0.68
	PC2	0.82	0.76
	PC3	0.81	0.75
Olmov2-Base	PC1	0.75	0.69
	PC2	0.86	0.79
	PC3	0.77	0.72
Olmov2-Instruct	PC1	0.74	0.68
	PC2	0.78	0.72
	PC3	0.77	0.71
Ministral	PC1	0.76	0.69
	PC2	0.84	0.77
	PC3	0.79	0.71

Table 7: Correlation between emotion ranking in latent space across models. Each model latent space's emotion ranking in this table is being correlated to Llama3.1-8B-Base.

H EMOTIONAL STEERING PER DATASET RESULTS

In Section 4.4, the results of the steering method were presented as averages across English and non-English datasets. In this appendix, the corresponding per-dataset results are presented. Table 8 shows that the learned steering approach achieves consistent and accurate control over internal emotional representations across a diverse set of languages and datasets for all three models. For most emotions, post-steering accuracy typically exceeds 85% across models. Performance is robust even in multilingual settings, with particularly strong results in French, German, and Italian. Steerability remains high for many emotions in Hindi, a lower-resource language, suggesting that lexical sparsity and data imbalance remain limiting factors for certain emotions in under-resourced settings across models. The semantic similarity loss across all datasets, models, and emotions is low, indicating that steering preserves much of the original representational structure while enabling control of emotional perception. LLaMA-3.1-8B is the most steerable model for a plurality of emotions and datasets; however, Olmov2 shows the greatest average delta across emotions and datasets—an outcome notable given that Olmov2-Instruct was previously shown in Section 4 to struggle with representing emotions in a unified manner. Appendix I reports ablations isolating the method's key factors.